Mathematical Modeling

Sheet 1: Primal Simplex, Alternative Optima, and Growth Rays

Personal study notes

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1 The task and the meaning of a simplex step

Rozwiąż podstawowym algorytmem sympleks. O ile to możliwe wystartuj z punktu x_0 Czy są rozwiązania alternatywne? W przypadku nieograniczonej funkcji podaj promień wzrostu. All six problems have three original decision variables. They are continuous real quantities; no physical units are supplied. Every stated nonnegativity condition is part of feasibility. For inequalities of the form $a^T x \le b$, write $a^T x + s = b$ with $s = b - a^T x \ge 0$. For $a^T x \ge b$, write $a^T x - s = b$ with $s = a^T x - b \ge 0$. The different sign records which residual is nonnegative. Auxiliary variables have objective coefficient zero and are numbered after x_3 separately in each problem.

A basis B selects three independent columns of the equality matrix A. Set all nonbasic variables to zero and solve $A_B x_B = b$. This produces a basic feasible solution only when $x_B \ge 0$. A feasible point need not be basic, and an inadmissible supplied point cannot start primal simplex. The basis order is the order of the rows in each displayed tableau.

1.1 Why the signs and the ratio test work

Multiplying the equations by A_B^{-1} gives

$$x_B + \bar{A}_N x_N = \bar{b}, \qquad \bar{A} = A_B^{-1} A, \qquad \bar{b} = A_B^{-1} b.$$

These row operations preserve every solution of the equality system. They change its representation, not its decisions. Substitute the basic equations into the maximization objective:

$$z = c^T x = z_0 - q_N^T x_N, \qquad z_0 = c_B^T \bar{b}, \qquad q_j = c_B^T \bar{a}_j - c_j.$$

Thus $q_j < 0$ means that increasing nonbasic x_j increases the objective. The alternative convention $c_j - z_j$ is $-q_j$ and reverses this test. The bottom row of each tableau lists $z_j - c_j$; its RHS cell records the current objective z_0, rather than another equation coefficient. The c_B column gives the objective coefficient of each basic variable. The z_j row sums each constraint column with these weights; subtract the displayed c_j row to obtain $q_j = z_j - c_j$. The same weighted sum on the RHS gives z_0. In Phase I these are the auxiliary objective coefficients; in Phase II they are the original ones. Basic indicators are always zero and do not by themselves imply alternative optima.

When x_k enters at value $\theta \geq 0$, the basic variables become $x_B(\theta) = \bar{b} - \theta \bar{a}_k$. A row with $\bar{a}_{ik} > 0$ limits the step to $\bar{b}_i / \bar{a}_{ik}$, because its basic variable decreases. A row with a zero or negative coefficient supplies no upper bound. Ratios are calculated in the column to the right of the RHS; $\longrightarrow$ in the separate Out column marks the chosen leaving row. The indicator row is labelled $z_j - c_j$; q_j abbreviates this difference in formulas. Vertical rules separate c_B, the basic-variable labels, the coefficient columns, and the RHS. Variable-column headings keep their order across pivots; only the basic-variable labels change. Artificial columns are removed after Phase I without reordering the retained columns. The minimum of the positive-coefficient ratios makes one basic variable zero and identifies a leaving variable. We choose a most negative nonbasic indicator, breaking entering ties by smaller variable index, for the improving pivots below. Zero-indicator pivots are used afterwards to exhibit alternatives.

If the pivot is in row r, the complete augmented-row update is

$$R'_r = R_r / \bar{a}_{rk}, \qquad R'_i = R_i - \bar{a}_{ik} R'_r \quad (i \neq r).$$

This changes the entering column into an identity column. Recomputing q with the new basic costs gives $z'_0 = z_0 - q_k \theta$. If $q_k < 0$ but no row coefficient is positive, there is no leaving variable: θ can grow without bound. The full direction is $d_k = 1$, $d_B = -\bar{a}_k$, with all other nonbasic components zero. It satisfies $d \geq 0$, $Ad = 0$ and $c^T d = -q_k > 0$. Its first three coordinates give a growth ray in the original variables.

For a finite optimum, $q \geq 0$ gives the global bound $z = z_0 - q_N^T x_N \leq z_0$. To claim another optimizer from a zero indicator, we still need a positive feasible step that changes the original point. An explicit optimal segment supplies stronger evidence than merely reporting a different basis.

2 Problem 1(a): a unique finite maximum

Task statement.

$$\max(-x_1 + x_2 + 3x_3), \qquad \begin{cases} -4x_1 + 2x_2 - x_3 \leq 2, \\ 2x_1 - x_2 + x_3 \leq 1, \\ 3x_1 + x_3 \leq 5, \\ x_1, x_2, x_3 \geq 0. \end{cases}$$

Add slacks x_4, x_5, x_6 to the three rows. The origin is feasible, and its slack basis $B = (4, 5, 6)$ has basic values $(2, 1, 5)$.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_3 \uparrow$)	Out
	c_j	-1	1	3	0	0	0			
0	x_4	-4	2	-1	1	0	0	2	–	
0	x_5	2	-1	1	0	1	0	1	$1/1 = 1$	$\longrightarrow$
0	x_6	3	0	1	0	0	1	5	$5/1 = 5$	
	z_j	0	0	0	0	0	0	0		
	$z_j - c_j$	1	-1	-3	0	0	0	$z_0 = 0$		

Enter x_3, since $q_3 = -3$, and remove x_5 at $\theta = 1$. Keep row 2, add it to row 1, and subtract it from row 3. The objective increases by $3 \cdot 1$.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_2 \uparrow$)	Out
	c_j	-1	1	3	0	0	0			
0	x_4	-2	1	0	1	1	0	3	$3/1 = 3$	$\longrightarrow$
3	x_3	2	-1	1	0	1	0	1	–	
0	x_6	1	1	0	0	-1	1	4	$4/1 = 4$	
	z_j	6	-3	3	0	3	0	3		
	$z_j - c_j$	7	-4	0	0	3	0	$z_0 = 3$		

Now enter x_2, since $q_2 = -4$. With x_4 leaving at step 3, add row 1 to row 2 and subtract row 1 from row 3. The original point becomes $(0, 3, 4)$ and the value becomes 15.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_1 \uparrow$)	Out
	c_j	-1	1	3	0	0	0			
1	x_2	-2	1	0	1	1	0	3	–	
3	x_3	0	0	1	1	2	0	4	–	
0	x_6	3	0	0	-1	-2	1	1	$1/3$	$\longrightarrow$
	z_j	-2	1	3	4	7	0	15		
	$z_j - c_j$	-1	0	0	4	7	0	$z_0 = 15$		

Enter x_1, whose indicator is -1, and remove x_6 at $\theta = 1/3$. Divide row 3 by 3 and add twice the normalized row to row 1.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
	c_j	-1	1	3	0	0	0	
1	x_2	0	1	0	$\frac{1}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	$\frac{11}{3}$
3	x_3	0	0	1	1	2	0	4
-1	x_1	1	0	0	$-\frac{1}{3}$	$-\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
	z_j	-1	1	3	$\frac{11}{3}$	$\frac{19}{3}$	$\frac{1}{3}$	$\frac{46}{3}$
	$z_j - c_j$	0	0	0	$\frac{11}{3}$	$\frac{19}{3}$	$\frac{1}{3}$	$z_0 = \frac{46}{3}$

All indicators are nonnegative. The original optimizer and value are

$$x^* = \left(\tfrac{1}{3}, \tfrac{11}{3}, 4\right), \qquad z^* = \tfrac{46}{3}.$$

The original slacks vanish, so all three resource rows are tight. The final objective identity holds for every feasible point:

$$-x_1 + x_2 + 3x_3 = \tfrac{46}{3} - \tfrac{11}{3}x_4 - \tfrac{19}{3}x_5 - \tfrac{1}{3}x_6 \le \tfrac{46}{3}.$$

Equality forces $x_4 = x_5 = x_6 = 0$. Solving the three tight original rows yields the displayed point uniquely. There are no alternative optimizers.

3 Problem 1(b): an unbounded feasible set with a finite maximum

Task statement.

$$\max(2x_1 - x_2 + 3x_3), \qquad \begin{cases} -3x_1 - x_2 - 2x_3 \le 3, \\ 3x_1 + x_3 \le 2, \\ -x_2 + x_3 \le 1, \\ x_1, x_2, x_3 \ge 0. \end{cases}$$

Add slacks x_4, x_5, x_6. The first inequality is automatically satisfied by nonnegative original variables, but retaining it gives a harmless basic slack. Start at the origin with $B = (4, 5, 6)$.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_3 \uparrow$)	Out
	c_j	2	−1	3	0	0	0			
0	x_4	−3	−1	−2	1	0	0	3	–	
0	x_5	3	0	1	0	1	0	2	$2/1 = 2$	
0	x_6	0	−1	1	0	0	1	1	$1/1 = 1$	$\longrightarrow$
	z_j	0	0	0	0	0	0	0		
	$z_j - c_j$	−2	1	−3	0	0	0	$z_0 = 0$		

Enter x_3 and remove x_6 with step 1. Add twice row 3 to row 1 and subtract row 3 from row 2.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_1 \uparrow$)	Out
	c_j	2	−1	3	0	0	0			
0	x_4	−3	−3	0	1	0	2	5	–	
0	x_5	3	1	0	0	1	−1	1	1/3	$\longrightarrow$
3	x_3	0	−1	1	0	0	1	1	–	
	z_j	0	−3	3	0	0	3	3		
	$z_j - c_j$	−2	−2	0	0	0	3	$z_0 = 3$		

The indicators $q_1 = q_2 = -2$ tie; choose x_1 by smaller index. The step is 1/3. Divide row 2 by 3 and add three copies to row 1; x_5 leaves.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_2 \uparrow$)	Out
	c_j	2	−1	3	0	0	0			
0	x_4	0	−2	0	1	1	1	6	–	
2	x_1	1	$\frac{1}{3}$	0	0	$\frac{1}{3}$	$-\frac{1}{3}$	$\frac{1}{3}$	$(1/3)/(1/3) = 1$	$\longrightarrow$
3	x_3	0	−1	1	0	0	1	1	–	
	z_j	2	$-\frac{7}{3}$	3	0	$\frac{2}{3}$	$\frac{7}{3}$	$\frac{11}{3}$		
	$z_j - c_j$	0	$-\frac{4}{3}$	0	0	$\frac{2}{3}$	$\frac{7}{3}$	$z_0 = \frac{11}{3}$		

Enter x_2, since $q_2 = -4/3$, and remove x_1 with step 1. Multiply row 2 by 3, add twice that normalized row to row 1, and add it to row 3.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
	c_j	2	−1	3	0	0	0	
0	x_4	6	0	0	1	3	−1	8
−1	x_2	3	1	0	0	1	−1	1
3	x_3	3	0	1	0	1	0	2
	z_j	6	−1	3	0	2	1	5
	$z_j - c_j$	4	0	0	0	2	1	$z_0 = 5$

The point is $x^* = (0, 1, 2)$, with maximum $z^* = 5$ and slacks $(8, 0, 0)$. For all feasible points the final objective is

$$2x_1 - x_2 + 3x_3 = 5 - 4x_1 - 2x_5 - x_6 \leq 5.$$

Equality forces $x_1 = x_5 = x_6 = 0$. The second row then fixes $x_3 = 2$, and the third fixes $x_2 = 1$; the optimizer is unique. The zero indicator of basic x_4 supplies no alternative decision.

The feasible set itself is unbounded: $(0, t, 0)$ is feasible for every $t \geq 0$. Its objective is $-t$, so that ray cannot improve a maximization objective. Unboundedness of the feasible set does not imply an unbounded maximum.

4 Problem 1(c): an entire segment of optimal solutions

Task statement.

$$\max(-4x_1 + 2x_2 + 3x_3), \qquad \begin{cases} 2x_1 - x_2 + x_3 \le 2, \\ -6x_1 + 3x_2 - 2x_3 \le 2, \\ -7x_1 + 4x_2 - 3x_3 \le 2, \\ x_1, x_2, x_3 \ge 0. \end{cases}$$

Add slacks x_4, x_5, x_6 and start with $B = (4, 5, 6)$ at the origin.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_3 \uparrow$)	Out
	c_j	-4	2	3	0	0	0			
0	x_4	2	-1	1	1	0	0	2	$2/1 = 2$	$\longrightarrow$
0	x_5	-6	3	-2	0	1	0	2	–	
0	x_6	-7	4	-3	0	0	1	2	–	
	z_j	0	0	0	0	0	0	0		
	$z_j - c_j$	4	-2	-3	0	0	0	$z_0 = 0$		

Enter x_3 and remove x_4 at $\theta = 2$. Add twice row 1 to row 2 and three times row 1 to row 3.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_2 \uparrow$)	Out
	c_j	-4	2	3	0	0	0			
3	x_3	2	-1	1	1	0	0	2	–	
0	x_5	-2	1	0	2	1	0	6	$6/1 = 6$	$\longrightarrow$
0	x_6	-1	1	0	3	0	1	8	$8/1 = 8$	
	z_j	6	-3	3	3	0	0	6		
	$z_j - c_j$	10	-5	0	3	0	0	$z_0 = 6$		

Enter x_2, since $q_2 = -5$, and remove x_5 with step 6. Add row 2 to row 1 and subtract row 2 from row 3.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_1 \uparrow$)	Out
	c_j	-4	2	3	0	0	0			
3	x_3	0	0	1	3	1	0	8	–	
2	x_2	-2	1	0	2	1	0	6	–	
0	x_6	1	0	0	1	-1	1	2	$2/1 = 2$	$\longrightarrow$
	z_j	-4	2	3	13	5	0	36		
	$z_j - c_j$	0	0	0	13	5	0	$z_0 = 36$		

All indicators are nonnegative, so $(0, 6, 8)$ is optimal with value 36. Here x_1 is *nonbasic* with $q_1 = 0$. To test for an actual alternative, let it enter with step $\theta = 2 > 0$, removing x_6. The original point changes and the objective change is $-q_1\theta = 0$. Add twice row 3 to row 2; x_6 leaves.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
	c_j	-4	2	3	0	0	0	
3	x_3	0	0	1	3	1	0	8
2	x_2	0	1	0	4	-1	2	10
-4	x_1	1	0	0	1	-1	1	2
	z_j	-4	2	3	13	5	0	36
	$z_j - c_j$	0	0	0	13	5	0	$z_0 = 36$

The second endpoint is $(2, 10, 8)$, also of value 36. The global certificate and complete set of optimizers are

$$-4x_1 + 2x_2 + 3x_3 = 36 - 13x_4 - 5x_5 \le 36,$$
$$x(t) = (t, 6 + 2t, 8), \qquad 0 \le t \le 2. \tag{1}$$

Indeed, equality forces $x_4 = x_5 = 0$. Their tight original rows give $x_3 = 8$ and $x_2 = 6 + 2x_1$. The third slack is $x_6 = 2 - x_1$, so nonnegativity gives exactly $0 \le x_1 \le 2$. This derives every optimizer, including nonbasic interior points of the segment.

5 Problem 1(d): starting at the supplied point

Task statement.

$$\max(x_1 + 2x_2 + x_3), \qquad \begin{cases} 2x_2 + x_3 \ge 4, \\ x_2 + x_3 \le 10, \\ x_1 + 2x_2 + 2x_3 \le 8, \\ x_1, x_2, x_3 \ge 0, \end{cases} \qquad x_0 = (0, 2, 0).$$

For the first row subtract the surplus $x_4 = 2x_2 + x_3 - 4 \ge 0$; for the other rows add slacks x_5, x_6. The equality equations are

$$2x_2 + x_3 - x_4 = 4, \qquad x_2 + x_3 + x_5 = 10, \qquad x_1 + 2x_2 + 2x_3 + x_6 = 8.$$

The supplied point is feasible and its auxiliary values are $(0, 8, 4)$. Choose $B = (2, 5, 6)$: the columns are independent, and the positive basic values are $(2, 8, 4)$. No artificial variable is needed merely because a $\ge$ row occurs. Divide the first equation by 2, subtract the resulting row from row 2, and subtract twice it from row 3.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_1 \uparrow$)	Out
	c_j	1	2	1	0	0	0			
2	x_2	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	2	–	
0	x_5	0	0	$\frac{1}{2}$	$\frac{1}{2}$	1	0	8	–	
0	x_6	1	0	1	1	0	1	4	$4/1 = 4$	$\longrightarrow$
	z_j	0	2	1	-1	0	0	4		
	$z_j - c_j$	-1	0	0	-1	0	0	$z_0 = 4$		

The indicators of x_1 and surplus x_4 tie at -1; choose x_1. Remove x_6 at $\theta = 4$. That row already contains the entering identity column, so the constraint coefficients do not change; the basic label and objective do.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_4 \uparrow$)	Out
	c_j	1	2	1	0	0	0			
2	x_2	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	2	–	
0	x_5	0	0	$\frac{1}{2}$	$\frac{1}{2}$	1	0	8	$8/(1/2) = 16$	
1	x_1	1	0	1	1	0	1	4	$4/1 = 4$	$\longrightarrow$
	z_j	1	2	2	0	0	1	8		
	$z_j - c_j$	0	0	1	0	0	1	$z_0 = 8$		

The original point $(4, 2, 0)$ is optimal with value 8. Its objective identity is

$$x_1 + 2x_2 + x_3 = 8 - x_3 - x_6 \le 8.$$

To find an alternative, enter nonbasic x_4 with $q_4 = 0$. Remove x_1 with step 4. Add half row 3 to row 1 and subtract half row 3 from row 2.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
	c_j	1	2	1	0	0	0	
2	x_2	$\frac{1}{2}$	1	1	0	0	$\frac{1}{2}$	4
0	x_5	$-\frac{1}{2}$	0	0	0	1	$-\frac{1}{2}$	6
0	x_4	1	0	1	1	0	1	4
	z_j	1	2	2	0	0	1	8
	$z_j - c_j$	0	0	1	0	0	1	$z_0 = 8$

The other endpoint is $(0, 4, 0)$, still of value 8. For any optimum, the certificate forces $x_3 = x_6 = 0$, so $x_1 + 2x_2 = 8$. The first row requires $x_2 \geq 2$ and $x_1 \geq 0$ requires $x_2 \leq 4$. Therefore the complete optimal segment is

$$x(t) = (8 - 2t, t, 0), \qquad 2 \leq t \leq 4. \tag{2}$$

The second row remains slack throughout, since $x_5 = 10 - t > 0$.

6 Problem 1(e): reject the supplied start and exhibit a growth ray

Task statement.

$$\max(2x_1 + x_2 + x_3), \qquad \begin{cases} 2x_1 + 3x_2 - x_3 \leq 9, \\ 2x_2 + x_3 \geq 4, \\ x_1 = 6, \\ x_1, x_2, x_3 \geq 0, \end{cases} \qquad x_0 = (0, 0, 6).$$

The supplied x_0 violates the equality $x_1 = 6$, although it satisfies the two inequalities and non-negativity. It cannot start primal simplex. Keep the equality and find a feasible point for the stated problem: with $x_1 = 6$ and $x_2 = 0$, the inequalities become $x_3 \geq 3$ and $x_3 \geq 4$. Choose $x_3 = 4$.

Add first-row slack x_4 and subtract second-row surplus x_5:

$$2x_1 + 3x_2 - x_3 + x_4 = 9, \qquad 2x_2 + x_3 - x_5 = 4, \qquad x_1 = 6.$$

The feasible augmented point is $(6, 0, 4, 1, 0)$. The independent columns $B = (4, 3, 1)$ give basic values $(1, 4, 6)$. Subtract twice the equality row from row 1, then add row 2 to row 1:

c_B	Basic	x_1	x_2	x_3	x_4	x_5	RHS	Ratio ($x_5 \uparrow$)	Out
	c_j	2	1	1	0	0			
0	x_4	0	5	0	1	-1	1	–	
1	x_3	0	2	1	0	-1	4	–	
2	x_1	1	0	0	0	0	6	–	
	z_j	2	2	1	0	-1	16		
	$z_j - c_j$	0	1	0	0	-1	$z_0 = 16$		

Only $q_5 = -1$ is negative. Its column is $(-1, -1, 0)^T$, so increasing x_5 *increases* the first two basic values and leaves x_1 unchanged. There is no positive coefficient and no leaving variable. The augmented direction and original growth ray are

$$d = (0, 0, 1, 1, 1), \qquad x(\theta) = (6, 0, 4) + \theta(0, 0, 1), \qquad \theta \geq 0.$$

For every $\theta \geq 0$, the original left sides are $8 - \theta \leq 9$, $4 + \theta \geq 4$ and $x_1 = 6$. The objective is $16 + \theta \to +\infty$. Thus the problem is feasible and unbounded above; it has no finite optimum whose alternatives could be classified. Rejection of the supplied point is different from infeasibility of the problem.

7 Problem 1(f): two supplied starts, one unbounded problem

Task statement.

$$\max(6x_1 + 2x_2 - 3x_3), \qquad \begin{cases} x_1 + 2x_2 - 3x_3 \le 3, \\ 2x_1 + 5x_2 - 5x_3 \le 7, \\ 2x_1 - 3x_2 - 7x_3 \le 8, \\ x_1, x_2, x_3 \ge 0. \end{cases}$$

i) $x_0 = (0, 1, 0)$, ii) $x_0 = (3, 0, 0)$, Add slacks x_4, x_5, x_6 to the three rows. The different supplied points require different initialization, even though the LP itself is identical.

7.1 Start i: a feasible point that is not basic

At $(0, 1, 0)$ the original row values are $(2, 5, -3)$ and the slacks are $(1, 2, 11)$. All constraints are satisfied. However, x_2, x_4, x_5, x_6 are positive: four positive augmented coordinates cannot fit into a basis of three columns, whose nonbasic coordinates must all be zero. No choice of basis makes this point a basic solution in this slack formulation.

We can use the supplied point to reach a feasible vertex before performing primal simplex. Keep $x_1 = x_3 = 0$ and increase $x_2 = 1 + \tau$. The slacks become

$$x_4 = 1 - 2\tau, \qquad x_5 = 2 - 5\tau, \qquad x_6 = 11 + 3\tau.$$

Feasibility requires $0 \le \tau \le \min(1/2, 2/5) = 2/5$. At the first boundary, $x_2 = 7/5$ and the slacks are $(1/5, 0, 61/5)$. The objective rises from 2 to 14/5. This preliminary move is a feasible initialization from a nonbasic point, not a simplex pivot from an existing basis. Now $B = (4, 2, 6)$ is independent and has three positive basic values. Divide original row 2 by 5, subtract twice the normalized row from row 1, and add three times it to row 3:

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_1 \uparrow$)	Out
	c_j	6	2	−3	0	0	0			
0	x_4	$\frac{1}{5}$	0	−1	1	$-\frac{2}{5}$	0	$\frac{1}{5}$	$(1/5)/(1/5) = 1$	$\longrightarrow$
2	x_2	$\frac{2}{5}$	1	−1	0	$\frac{1}{5}$	0	$\frac{7}{5}$	$(7/5)/(2/5) = 7/2$	
0	x_6	$\frac{16}{5}$	0	−10	0	$\frac{3}{5}$	1	$\frac{61}{5}$	$(61/5)/(16/5) = 61/16$	
	z_j	$\frac{4}{5}$	2	−2	0	$\frac{2}{5}$	0	$\frac{14}{5}$		
	$z_j - c_j$	$-\frac{26}{5}$	0	1	0	$\frac{2}{5}$	0	$z_0 = \frac{14}{5}$		

Enter x_1 with $q_1 = -26/5$ and remove x_4 at $\theta = 1$. Multiply row 1 by 5, subtract (2/5) times that normalized row from row 2, and subtract (16/5) times it from row 3.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_3 \uparrow$)	Out
	c_j	6	2	−3	0	0	0			
6	x_1	1	0	−5	5	−2	0	1	–	
2	x_2	0	1	1	−2	1	0	1	$1/1 = 1$	$\longrightarrow$
0	x_6	0	0	6	−16	7	1	9	$9/6 = 3/2$	
	z_j	6	2	−28	26	−10	0	8		
	$z_j - c_j$	0	0	−25	26	−10	0	$z_0 = 8$		

The point is $(1, 1, 0)$ and the objective is 8. Enter x_3 with $q_3 = -25$, and remove x_2 with step 1. Add five copies of row 2 to row 1 and subtract six copies from row 3:

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_4 \uparrow$)	Out
	c_j	6	2	−3	0	0	0			
6	x_1	1	5	0	−5	3	0	6	–	
−3	x_3	0	1	1	−2	1	0	1	–	
0	x_6	0	−6	0	−4	1	1	3	–	
	z_j	6	27	−3	−24	15	0	33		
	$z_j - c_j$	0	25	0	−24	15	0	$z_0 = 33$		

This gives $(6, 0, 1)$, value 33. The negative indicator $q_4 = -24$ means this point is not optimal. The next entering variable is the slack x_4, whose whole column is nonpositive. The growth ray is derived in Section 7.3.

7.2 Start ii: the supplied point is already basic

At $(3, 0, 0)$ the slacks are $(0, 1, 2)$. Columns $B = (1, 5, 6)$ form an independent basis with positive values $(3, 1, 2)$. Keep original row 1, subtract twice it from rows 2 and 3, and recompute the objective using the basic cost of x_1:

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_3 \uparrow$)	Out
	c_j	6	2	−3	0	0	0			
6	x_1	1	2	−3	1	0	0	3	–	
0	x_5	0	1	1	−2	1	0	1	$1/1 = 1$	$\longrightarrow$
0	x_6	0	−7	−1	−2	0	1	2	–	
	z_j	6	12	−18	6	0	0	18		
	$z_j - c_j$	0	10	−15	6	0	0	$z_0 = 18$		

Enter x_3, whose indicator is -15. Remove x_5 at $\theta = 1$. Add three copies of row 2 to row 1 and add row 2 to row 3.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_4 \uparrow$)	Out
	c_j	6	2	−3	0	0	0			
6	x_1	1	5	0	−5	3	0	6	–	
−3	x_3	0	1	1	−2	1	0	1	–	
0	x_6	0	−6	0	−4	1	1	3	–	
	z_j	6	27	−3	−24	15	0	33		
	$z_j - c_j$	0	25	0	−24	15	0	$z_0 = 33$		

The new basis is $(1, 3, 6)$, exactly the same canonical system reached from start i. The original point is $(6, 0, 1)$ and its objective is 33. The two different histories therefore lead to the same unboundedness test.

7.3 A ray, not a last optimal point

At the shared final basis, $q_4 = -24$ and $\bar{a}_4 = (-5, -2, -4)^T$. Increasing nonbasic slack x_4 by θ increases basic (x_1, x_3, x_6) by $(5, 2, 4)\theta$. No basic value decreases, so no finite ratio test bounds the step. The full augmented direction and its original-variable projection are

$$d = (5, 0, 2, 1, 0, 4), \qquad x(\theta) = (6, 0, 1) + \theta(5, 0, 2), \qquad \theta \geq 0.$$

The three original row values on this ray are

$$3 - \theta \leq 3, \qquad 7 \leq 7, \qquad 5 - 4\theta \leq 8.$$

Nonnegativity holds for every $\theta \geq 0$ and

$$6x_1(\theta) + 2x_2(\theta) - 3x_3(\theta) = 33 + 24\theta \longrightarrow +\infty.$$

The negative objective coefficient of x_3 does not prevent improvement: its increase permits a sufficiently profitable increase in x_1. A slack can enter a basis just like an original variable; projecting the resulting direction explains the change in original decisions. There is no finite optimal solution for either start.

8 What the six outcomes teach

Problem	Outcome	Original solution or family
1(a)	Unique maximum 46/3	$(1/3, 11/3, 4)$
1(b)	Unique maximum 5	$(0, 1, 2)$
1(c)	Maximum 36, alternatives	$(t, 6+2t, 8)$, $0 \leq t \leq 2$
1(d)	Maximum 8, alternatives	$(8-2t, t, 0)$, $2 \leq t \leq 4$
1(e)	Unbounded above	$(6, 0, 4) + \theta(0, 0, 1)$
1(f)	Unbounded above	$(6, 0, 1) + \theta(5, 0, 2)$

For the two rays, $\theta \geq 0$. The growth rate depends on the scaling of the direction; the existence of a positive rate is the unboundedness certificate. Check domains and all original constraints before choosing a start. Feasibility alone does not establish basicness. At a finite optimum use the indicator identity to prove a global bound, then derive uniqueness or the entire optimal face. For an unbounded problem provide both a feasible base point and a direction; a failed ratio test without that interpretation is an incomplete explanation.

9 Sources and editorial remarks

Task source: K. Junosza-Szaniawski, `../execrices-2026-27/ProgMatc1.pdf`, PDF page 1, task 1(a)–(f). The original Polish common instruction, mathematical data, supplied points and variable domains are preserved. The stated equality $x_1 = 6$ in 1(e) is retained; the supplied point fails that equality. The wording "if possible" permits rejecting the start. No undocumented change to the problem is assumed. Both source pages were inspected visually to confirm the statements.

Editorial additions and alignment. The hand derivations, explicit basis tests, full tableaus, complete optimal segments and growth-ray certificates are added explanations. All tableaus and pivot row operations were independently checked using exact rational arithmetic. The convention $q = z - c$, slack/surplus meanings, and certificate arguments match Chapters 2–4 of the reference notes. These are current supplied exercise tasks; their presence establishes task coverage, not a claim about how much was taught or explained in a particular class session.

MATLAB comparison. The actual companion is the primal-simplex MATLAB script:

`matlab/exercises/sheet_01_primal_simplex.m`

Run it from the repository root:

```
run('matlab/exercises/sheet_01_primal_simplex.m')
```

It uses base MATLAB, fixed supplied matrices and the prescribed pivot paths; it does not call an optimization solver or select pivots automatically. It recomputes each basis with backslash, checks every displayed tableau, indicator, point, objective change and row operation, tests the supplied

starts, and verifies the finite bounds, optimal-segment endpoints and ray equations. The absolute tolerance is 10^{-10} for these small dimensionless problems. Exact hand fractions are compared with floating-point calculations on identical inputs. Endpoint and direction identities prove the entire segments/rays; additional sample points are numerical checks rather than substitutes for proof.

Execution status: passed in MATLAB. The user supplied output on 2026-10-11 and confirmed version 26.2.0.3386108 (R2026b). All prescribed paths and certificate assertions passed. The largest checked tableau/basis discrepancy was $7.11 \cdot 10^{-15}$, below the absolute tolerance 10^{-10}; this is an observed discrepancy for these checks, not a universal solution-error bound. Independent exact-arithmetic verification establishes the hand calculations. The supplied output is retained in `docs/matlab-verification.md`.